

Atomic Structure

Atom: Atoms are made of protons, neutrons and electrons.

Description of three permanent fundamental particles

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(1) **Electron:** In 1897, while experimenting on cathode rays. Sir J. J. Thomson discovered electron. Electron is the smallest particle. (i) The mass of an electron is 9.1085×10^{-28} g, which is $\frac{1}{1837}$ part of the mass of a proton, (ii) Electron is negatively charged and the amount of its charge is -1.6×10^{-19} C (columb). (iii) The symbol for electron is 'e'. (iv) Electrons move outside the nucleus of an atom.

(2) **Proton:** In 1919 Rutherford proved that like electron proton is also present in atoms of all elements, (i) The mass of a proton is 1.673×10^{-24} g, which is almost same as that of hydrogen atom. In atomic mass scale it is equal to 1.007276 amu. So its mass is generally taken as one unit, (ii) Proton is positively charged and its amount is equal to that of an electron. (iii) The symbol for proton is (small) 'p'. (iv) The protons remain in the nucleus of an atom.

(3) **Neutron:** Like electron and proton, neutron is also a fundamental particle. In 1920 Rutherford first postulated the existence of neutron. In 1932 Chadwick discovered neutron, (i) The mass of a neutron is 1.675×10^{-24} g. (ii) Neutron is uncharged, (iii) The symbol for neutron is (small) 'n'. (iv) Neutrons reside inside the nucleus of an atom. Except 'Hydrogen-1 isotope' neutron is present in all atoms.

Atomic Number and Atomic Mass Number

(1) Atomic number

Definition : The number of protons in the nucleus of an atom of an element is called 'Atomic number' of that element. It is denoted by 'Z'. For example there are 11 protons in the nucleus of a sodium atom. Hence the atomic number of sodium is 11.

Atomic number = number of protons

$$\text{i.e. } Z = p$$

(2) Atomic mass number

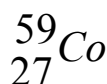
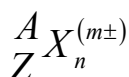
Definition: The total number of protons and neutrons in the nucleus of an atom is known as 'Atomic Mass Number'. It is denoted by 'A'. For example generally the nucleus of sodium atom contains 11 protons and 12 neutrons. Hence the atomic mass number of those sodium atoms is $(11 + 12) = 23$.

Atomic mass number = number of protons + number of neutrons

$$\text{i.e. } A = p + n$$

(3) Rule for Writing atomic number, mass number, charge and no. of atoms:

The atomic number Z is written on the left side of the symbol of the element as subscript, the mass number A is written on the left side of the symbol as superscript. The charge ($m \pm$) is written on the right side of the symbol as superscript. The number of atoms (n) in a molecule/ion is written on the right side of the symbol as subscript. If the symbol of the element be X, then all the rules may be presented as follows:



Isotope

The masses of atoms of the same element may be different. For example the masses of oxygen atoms in nature are 16, 17 and 18 units, these are known as isotopes.

Definition : The atoms, which have same number of protons, but whose masses are different, are called isotopes to each other. Isotopes are atoms of same element. They have same position in the periodic table. Hence this name 'Isotope' is attributed to them. 'Iso' means same and 'topos' means position, i.e. position in Periodic table.

For example, there are three kinds of carbon atom, called carbon-12, carbon-13 and carbon-14. They all have the same number of protons, but the no. of neutrons varies. These different atoms are called **isotopes**.

${}^{12}_6\text{C}$	${}^{13}_6\text{C}$	${}^{14}_6\text{C}$
carbon-12	carbon-13	carbon-14
6 protons	6 protons	6 protons
6 neutrons	7 neutrons	8 neutrons

Isobar and Isotone

Definition of isobar: The atoms whose mass numbers are equal, but the atomic numbers are different are called isobars to one another. This means that the proton numbers are different, i.e., they are atoms of different elements. So their chemical and physical properties are different. For example -

${}^{64}_{28}\text{Ni}$	${}^{64}_{29}\text{Cu}$	${}^{64}_{30}\text{Zn}$
Nickel	Copper	Zinc

Definition of isotone : The atoms whose neutron numbers are same, but the atomic numbers and mass numbers are different are called isotones to one another. Since isotones are atoms of different elements, their chemical and physical properties are different.

${}^{30}_{14}\text{Si}$	${}^{31}_{15}\text{P}$	${}^{32}_{16}\text{S}$
Silicon	Phosphorus	Sulphur

Mass of an atom and Relative atomic mass

Atoms are so tiny that they cannot be seen even by most powerful microscope and hence they can not be directly weighed. But scientists have determined their masses indirectly. For example.

mass of one (Hydrogen-1) atom = 1.673×10^{-24} g

mass of one (Carbon-12) atom = 1.9924×10^{-23} g

mass of one (Oxygen-16) atom = 2.6560×10^{-23} g

Since these masses are very small, a new unit is needed to express these masses. In 1961 IUPAC, an international association adopted a new unit known as **atomic mass unit**, in brief **amu**.

Definition of atomic mass unit: One atomic mass unit is defined as $\frac{1}{12}$ th part of mass of one

${}^{12}_6\text{C}$ atom.

$$\therefore 1 \text{ amu} = \frac{\text{mass of one } {}^{12}\text{C atom}}{12} = \frac{1.9924 \times 10^{-23}}{12} \text{ g} = 1.66 \times 10^{-24} \text{ g}$$

Relative atomic mass is the ratio of the mass of an atom to that of this atomic mass unit (amu). Hence it is a pure number. It is also called **atomic mass**.

For example relative atomic mass or atomic mass of Hydrogen is 1.008, that of (Oxygen-16) is

15.9998, that of (C-12) is 12.000. In nature almost all elements are composed of several isotopes. So the average relative atomic mass of each element is used to express the atomic mass of that element. For example atomic mass of carbon is 12.011.

Atomic Models:

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After the discovery of electrons the realization was that atoms are divisible. Different Scientists forwarded different concepts about the structure of atom in the period of 1897-1932. These concepts are known as atomic models. The atomic models are :

1. Thomson atom model : forwarded in 1898
2. Rutherford atom model : forwarded in 1911
3. Bohr atom model : forwarded in 1913
4. Bohr-Sommerfield atom model : forwarded in 1916
5. Wave-mechanical atom model : forwarded in 1924

Q. Describe briefly Rutherford's atom model with its limitations.

Answer:

Rutherford Atom Model

Based on α -particle scattering experiment. Rutherford forwarded his atom model. The main postulates of Rutherford atom model are as follows:

(1) **An atom has two parts:** (i) nucleus and (ii) extra nuclear part. Nucleus is a minute, massive positively charged body at the centre of the atom. All the positive charges and almost total mass of the atom are present in the nucleus. The volume of the nucleus is very small compared to that of the atom as a whole.

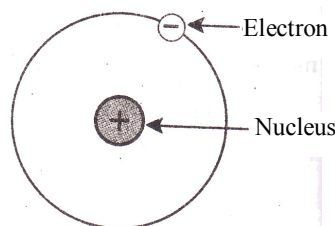


Fig. 1: Rutherford atom model

(2) **Atom is electrically neutral:** So a number of electrons equal to positive charges of the nucleus are present in an atom. Electrons are in constant motion around the nucleus in different orbits like the planets in a solar system.

(3) Due to the constant motion of the electrons around the nucleus, the outward centrifugal force is exactly counter-balanced by the inward electrostatic attraction between the electrons and nucleus i.e. **centripetal force**.

Since Rutherford atom model resembles the solar system, it is also known as **solar system atom model**. Again since the concept of nucleus was first introduced in this model, it is also known as **nuclear model of atom**.

Limitations of Rutherford Atom Model:

There are some limitations in Rutherford atom model, which are stated below:

(1) The model has been compared with the solar system. The rotating electrons are negatively charged and repel each other while the rotating planets are neutral and have force of gravitation acting between them. On the otherhand the force between nucleus and electrons is electrostatic. So comparison between gravitational force and electrostatic force is not correct.

(2) According to Maxwell's electromagnetic theory, any charged particle while revolving will emit radiation and lose energy. Thus rotating

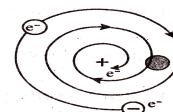


Fig. 2: Spiral path

electrons round the nucleus will lose energy and its orbit will become smaller and smaller and it will ultimately fall into the nucleus following a spiral path in about 10^{-8} second. But it does not occur practically.

(3) This model can not give any explanation of the atomic spectra.

(4) Rutherford did not give any idea about the shape and size of the orbits of electrons in an atom.

(5) There was no explanation about the rotation of many electrons in an orbit.

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Q. Describe briefly Bohr's atom model with its limitations.

Answer:

Bohr Atom Model

In 1913 Neils Bohr proposed his famous atom model, which not only described the structure of atom, but also gave an explanation about the origin of atomic spectra. **The postulates of Bohr Atom model can be divided into three important parts as follows:**

(1) Concept of stationary orbit or energy level: According to Bohr, electrons can rotate in definite circular paths around the nucleus. During the rotation the electrons will follow all rules of physics. But so long electron or electrons remain in a definite orbit; it will not absorb or radiate any energy. These circular paths are known as stationary orbits or **energy levels of electrons.**

Each energy level has a definite or fixed energy. These energy levels are designated by n . For the nearest energy level or orbit $n = 1$, for second orbit $n = 2$ and so on.

(2) Concept of angular momentum of electrons: The permissible orbits or energy levels are those for which angular momentum of the revolving electrons are an integral multiple of $\frac{h}{2\pi}$. The angular momentum, $mvr = n \times \frac{h}{2\pi}$; where m = mass of an electron, v = linear velocity of electron, r = radius of orbit, $n = 1, 2, 3 \dots$ represents the number of orbit, h = Planck's constant.

(3) Concept of origin of spectral lines : An electron can jump from an orbit or energy level to another. When electron jumps from higher energy level (E_2) to lower energy level (E_1), then it emits energy. On the other hand if an electron jumps from lower energy level (E_1) to higher energy level (E_2), then it absorbs energy. The energy emitted or absorbed is the difference between the energies of the two levels such that $(E_2 - E_1) = h\gamma$; where γ is the frequency of emitted or absorbed energy and h is the plank's constant. Here the related radiation energy, $\Delta E = (E_2 - E_1)$ will produce an atomic spectral line corresponding to frequency γ (neu).

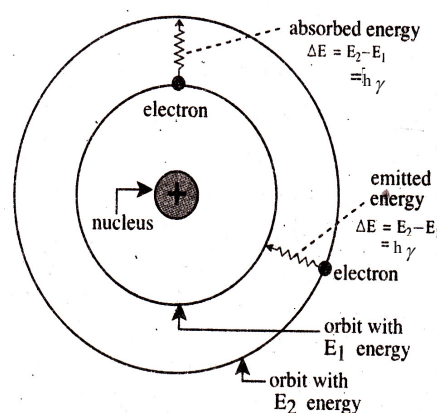


Fig. 3: Bohr atom model

Limitations of Bohr Atom Model:

- (1) Bohr atomic model can explain the spectral lines of H-atom and hydrogen like ions of one electron e.g. He^+ , Li^{2+} , it can not explain the spectral lines of atoms with many electrons.
- (2) According to Bohr model when an electron jumps from one energy level to another level,

a single line in spectrum is supposed to appear. However, when a spectrograph is developed with resolving power, there are two or more lines very close together are observed. Bohr atom model can not give any explanation on this **multiple lines in atomic spectrum**.

(3) According to Heigenberg's uncertainty principle the exact position in orbit and momentum of a very small particle like electron can not be determined at the same time with good precision. But in Bohr atom model that has been done.

(4) According to Bohr atom model electrons revolve in circular orbits but modern research reveals that electrons revolve three dimensionally.

(5) Bohr atom model fails to recognize the wave property of revolving electrons which was established by De Broglie.

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Explanation of atomic spectrum of H-atom according to Bohr atom model

According to Bohr theory the electrons in infinite number of excited hydrogen atoms in a sample absorb different amount of energy from same source and jump to different levels of higher energy. These states are unstable and so the electrons jump back to lower energy levels and thereby emit electromagnetic radiation of different energies and hence of different but definite frequencies. When an electron jumps from n_2 th energy level to n_1 th energy level, then the wave number of emitted radiation is given by the following equation :

$$\bar{\nu} = R \times \left(\frac{1}{n_2^2} - \frac{1}{n_1^2} \right)$$

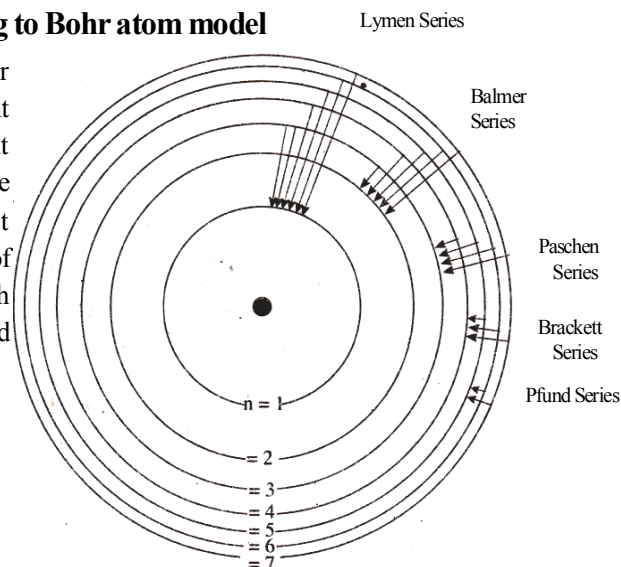


Figure: Atomic spectrum of H-atoms

Lyman series arises, when electrons jump from higher levels, $n_2 = 2, 3, 4$ to $n_1 = 1$ level

Balmer series arises, when electrons jump from higher levels, $n_2 = 3, 4, 5$ to $n_1 = 2$ level

Paschen series arises, when electrons jump from higher levels, $n_2 = 4, 5, 6$ to $n_1 = 3$ level

Brackett series arises, when electrons jump from higher levels, $n_2 = 5, 6, 7$ to $n_1 = 4$ level

Pfund series arises, when electrons jump from higher levels, $n_2 = 6, 7, 8$ to $n_1 = 5$ level

The formation of different lines in the atomic spectrum of hydrogen is shown in Fig.

Wave Mechanical Model of Atom

(a) Relationship between moving particle and wave

In Bohr atom model electron is treated only as a moving particle. In 1924 Louis de Broglie put forward a theory that moving electrons possess the characteristics of both particle and wave. According to Plank's quantum theory the energy of a photon, $E = h\gamma$. On the other hand according to Einstein's theory of equivalence of mass and energy, $E = mc^2$, where m = mass of a moving particle, E = equivalent energy and c = velocity of light.

Combining these two equations we get:

$$mc^2 = h\gamma = \frac{hc}{\lambda}; [\because c = \gamma\lambda]; \therefore mc = \frac{h}{\lambda}$$

Since mass of a photon = m, velocity of a photon = c, so m x c = momentum of the photon.

$$\therefore \text{momentum}, m \times c = \frac{h}{\lambda}; \text{or}, \lambda = \frac{h}{\text{momentum}(m \times c)}$$

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Broglie said that wave is associated also with a moving particle, the wave length of such 'matter wave' follows similar equation.

Since for matter, momentum p = mv, the equation for 'matter wave' becomes,

$$\lambda = \frac{h}{mv} \text{ Where } \lambda = \text{wave length of 'matter wave', } m = \text{mass of the particle, } v = \text{velocity of the particle.}$$

This equation is known as **De Broglie equation**.

(b) Explanation of a postulate of Bohr atom model with De-Broglie equation. ,

According to De Broglie, electrons move around the nucleus as wave. In that case, the wave should be stationary orbit (fig). Then the circumference of the orbit must be an integral multiple of wave length of moving electron wave. (λ).

$$\text{i.e. } 2\pi r = n \times \lambda = n \times \frac{h}{mv}; \therefore \lambda = \frac{h}{mv}$$

$$\text{or, } mvr = n \times \frac{h}{2\pi}$$

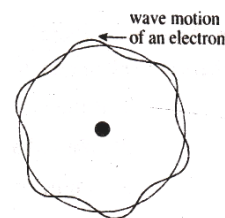


Fig. : Wave motion of electron in its orbit

Where, mVr = angular momentum of the moving electron.

This is the second postulate of Bohr atom model. Before De-Broglie's theory, no explanation could be forwarded for this equation of Bohr atom model.

(c) Heisenberg's Uncertainty, Principle

From the particle and wave nature of electrons Heisenberg proved mathematically that it is impossible to measure the exact velocity and exact position of an electron simultaneously. If Δx is the uncertainty in position of the moving particle and Δp is the uncertainty in momentum of the particle, then

$\Delta x \times \Delta p \geq h$ where h = Planck's constant. From this equation it is evident that if we want to determine the accurate position of an electron, then its velocity becomes uncertain and vice versa.

(d) Schrodinger's Wave Equation

To explain the dual properties (wave and particle) of electrons, Schrödinger came forward with his theory of wave mechanics in 1926. He established a three dimensional general mechanics equation called Schrödinger's wave equation. The equation is expressed as follows:

$$\frac{\delta^2\psi}{\delta x^2} + \frac{\delta^2\psi}{\delta y^2} + \frac{\delta^2\psi}{\delta z^2} + \frac{8\pi^2 m}{h} \times (E - \mu)\psi = 0$$

This equation is valid for all moving particles including electron. Here Ψ (psi) is the amplitude of the wave in space (x, y, z); m = mass of the particle, E = total energy of the particle; μ = potential energy of the particle; h = Planck's constant. On solving the equation, we can get different values of E and for every value of E, several equations for Ψ . Of them only some values satisfy some special conditions, they are called **eigen values** (i.e. acceptable) and the values of wave function Ψ correspond to these value of E are **eigen functions**. The eigen function actually relates to atomic orbital. The region in space where there is maximum probability of finding the electron-charge cloud is called **orbital**. Since a large number of solutions for Ψ are possible, hence to obtain a meaningful solution, three integers n, l, m were introduced. These three integers n, l, m are called **quantum numbers**. Thus each wave function Ψ describes a possible state of the electron in an atom.

(e) Physical significance of Ψ^2

Here Ψ^2 expresses the probability of finding electron at a place. Again where the probability of finding the electron is high, the electron cloud density will also be high. So Ψ^2 also expresses the electron cloud density at a place. Again the total probability of finding the electron must be unity. So

$$\int_0^\infty \Psi^2 dv = 1$$

Orbit and orbital

1. Orbit: According to Bohr atom model there are some fixed circular paths for electrons to revolve around the nucleus. These fixed circular paths are known as orbits. The orbits are characterized by values of principal quantum number. The orbit nearest to the nucleus is the first orbit and for it n = 1.

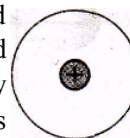


Fig. \ : Orbit of electron in an atom

Similarly the next orbits are known as 2nd, 3rd and 4th orbits and for them n = 2, 3, 4 etc. They are gradually more apart from the nucleus and hence their sizes increase gradually in this order.

2. Orbital: An atomic orbital is defined as the region around the nucleus in which the probability (90%) of finding the electron cloud is maximum.

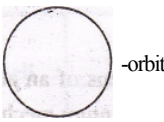
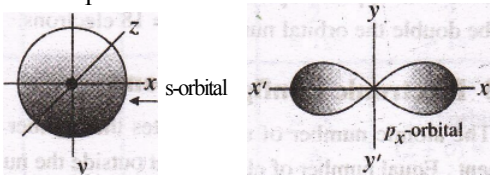
The probability (Ψ^2) of finding an electron is not the same every where in an atom. Therefore the density of the electron cloud is also not the same every where in an atom. The electron cloud being denser in some regions and less dense in some other region, the probability of finding an electron cloud decreases with distance, but does not become zero. Therefore, an orbital does not have a rigid boundary. In photograph a rotating electron appears as a cloud.



Fig. : s-orbital of electron in atom

Differences between Orbit and Orbital

Table: Comparison between Orbits and Orbital

Orbit	Orbital
1. The word 'orbit' originates from Bohr atom model. According to this model, electrons revolve around nucleus in definite circular paths. These paths are called orbit.	1. The word 'orbital' originates from quantum mechanics. The regions around the nucleus, where the probability of finding electrons is high (about 90%), are known as orbital. This word also means definite energy level in an atom.
2. The orbits are circular. The first orbit is nearest to the nucleus and 2nd, 3rd, 4th orbits etc. are apart more from nucleus gradually.  Fig (a) : orbit	2. The shapes of different orbitals are different for example s-orbitals are spherical, p-orbitals are dumbbell shaped with two lobes, d-orbitals are double dumbbell shaped with four lobes.  Fig (b) : orbitals
3. Orbits are related to principal quantum number, n. Here, n = 1 and 2 indicate 1st and 2nd orbits.	3. Orbitals are related to three quantum numbers n, l and m. For example 3p _x , 3p _y , 3p _z ,
4. Orbits are designated by the letters K, L, M, N, O etc. When n = 1, it is K orbit; when n = 2, it is L orbit and so on.	4. When subsidiary quantum number l = 0, 1, 2, 3 etc., the orbitals are designated s, p, d, f etc. respectively. Here for precision principal quantum number and orientation mentioned. For example 2p _x , 2p _y , 2p _z etc,
5. Each orbit can accommodate maximum 2n ² electrons; where n = principal quantum number. So K orbit can accommodate 2, L orbit 8, M orbit 18 and N orbit 32 electrons	5. Each orbital can accommodate maximum 2 electrons with opposite spin. For example one s orbital can accommodate 2 electrons, three p orbitals 6 electrons, five d orbitals 10 electrons and seven f orbitals 14 electrons.

2.19 Quantum Numbers

In an atom, the electrons are arranged in different energy levels. These electrons not only rotate around the nucleus but also rotate around their own axes. According to the quantum mechanics, three interrelated numbers are required to express the **size, shape and orientation of the orbitals** of electrons in an atom. Moreover, there is a number to express the direction of spin of the electron. These four correlated numbers are called quantum numbers which are as follows :

1. Principal quantum number, n
2. Azimuthal or subsidiary quantum number, l
3. Magnetic quantum number, m
4. Spin quantum number, s

1. Principal quantum number: The quantum number that expresses the main energy level in which the electrons are moving around the nucleus is the principal quantum number. It is expressed by n , where $n = 1, 2, 3, 4$ etc. It also represents the average distance of the electron's orbit from the nucleus. As the value of n increases, the distance of the orbit from the nucleus, i.e. the size of the orbit increases. Bohr gave the idea of this quantum number. When $n = 1, 2, 3, 4, 5$ etc. the energy levels are known as K, L, M, N, O shell respectively. The maximum number of electrons which can rotate in each particular orbit is given by $2n^2$.

Table : Different orbits or shells & their accommodated electrons

Orbit or shell in an atom	K	L	M	N	O	P	0
Principal quantum number, n	1	2	3	4	5	6	7
Total number of electrons, $2n^2$	2	8	18	32	50	72	98

2. Secondary quantum number: Each principal shell is divided into several subshells. The Quantum number which is used to indicate subshells is known as subsidiary quantum number. It is denoted by ' l ', it may have values from 0 to $(n - 1)$. The value of ' l ' determines the shape of the subshells or **orbitals**. When $l = 0, 1, 2, 3$, then the subshells are known as s, p, d and f orbitals respectively. The electrons rotating in these subshells are called s, p, d, f electrons respectively.

For 1st shell, $n = 1, l = 0$, i.e. 1st shell has 1 type orbital namely 1s

For 2nd shell, $n = 2, l = 0, 1$, i.e. 2nd shell has 2 types orbital namely 2s, 2p

For 3rd shell, $n = 3, l = 0, 1, 2$, i.e. 3rd shell has 3 types orbital namely 3s, 3p, 3d

For 4th shell, $n = 4, l = 0, 1, 2, 3$, i.e. 4th shell has 4 types orbital namely 4s, 4p, 4d, 4f

The maximum capacity of an orbital to hold electrons is $2 \times (2l + 1)$; where $l = 0, 1, 2, 3$ etc.

[The symbols for orbitals are taken as s for sharp, p for principal, d for diffused and f for fundamental corresponding to the four types of spectra obtained from alkali metals at the primary stage. In a particular orbit or shell, the various orbitals possess energy in the following order $s < p < d < f$.]

3. Magnetic quantum number: When a charged body moves around another charged body magnetic field is produced. In an atom, as negatively charged electrons move around the positively charged nucleus, a magnetic field is produced in an atom. As a result, each electron in an atom behaves as a tin magnet. The orbitals are responsible to magnetic moment and they are oriented in space. To describe the orientation, magnetic quantum number is introduced. It is denoted by ' m ' and its value depends on the value of ' l '. It may have value from $-l$ to $+l$ including zero. So there may be $(2l + 1)$ values of ' m ' for each value of ' l '. Its value gives the total number of orbitals present in each orbit.

For subshell s, $l = 0$ then $m = 0$ $\therefore$ in s subshell exists 1 orbital

For subshell p, $l = 1$ then $m = 1, 0, -1$ $\therefore$ in p subshell exist 3 orbitals

For subshell d, $l = 2$ then $m = 2, 1, 0, -1, -2$ $\therefore$ in d subshell exist 5 orbitals

For subshell f, $l = 3$ then $m = 3, 2, 1, 0, -1, -2, -3$ $\therefore$ in f subshell exist 7 orbitals

4. Spin quantum number: while rotating around the

nucleus the electrons also rotate around its own axis either in clockwise or in anticlockwise direction. Spin quantum number is used to describe this. It is expressed

by s and its value is $+\frac{1}{2}$ when clockwise and its value is $-\frac{1}{2}$ when

anti clockwise. This is also denoted by arrows pointing up and down

i.e. $\uparrow$ and $\downarrow$ respectively.

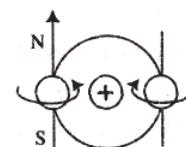


Fig.: Electrons' spin & magnetic field

Aufbau Principle

In an atom the electrons occupy different orbitals according to their increasing energy levels. That means that electrons first occupy the orbital of lowest energy and systematically then the orbitals of higher energy level.

This rule is called **Aufbau principle**. The word 'aufbau' is a German word, meaning 'building up'.

The orbital, which has lower value of $(n + l)$ has the lower energy and higher value of $(n + l)$ has higher energy. Here 'n' and 'l' are principal and subsidiary quantum numbers.

The orbitals of different energy from lower to higher gradually are as follows: 1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p, 6s, 4f, 5d, 6p, 7s

This gradational order can be remembered with the help of general scheme shown in Fig.

Determination of order of orbitals : To determine the comparative energy of different orbitals the following two rules are generally used :

Rule-1: Among several orbitals, the orbital whose $(n + l)$ value is less, will be of lower energy. For example among 3d, 4s, 4p orbitals :

for 3d orbital: $(n + l) = (3 + 2) = 5$

for 4s orbital: $(n + l) = (4 + 0) = 4$

for 5p orbital: $(n + l) = (5 + 1) = 6$

So among these three orbitals 4s orbital will be of lowest energy, the 3d will be of higher energy and 5p will be of highest energy content.

Rule-2 : If two or more orbitals possess same value of $(n + l)$, then the orbital whose value of 'n' is less, will be of lower energy. For example among 3d, 4p and 5s orbitals :

For 3d orbital: $(n + l) = (3 + 2) = 5$

for 4p orbital: $(n + l) = (4 + 1) = 5$

for 5s orbital: $(n + l) = (5 + 0) = 5$

For all these orbitals $(n + l)$ value is equal to 5. Since for 3d orbital the value of n is less than the others, it will be of lowest energy; then comes 4p orbital and then comes 5s orbital having highest energy. It must be remembered that these calculations are guide lines. Some times it is not strictly followed but it is a very useful rule.

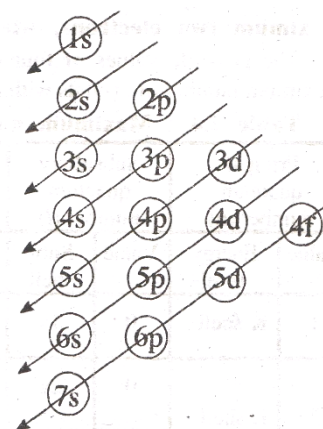
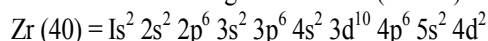


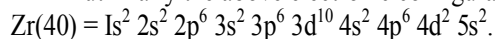
Fig. : Order of energy level of different orbitals

Rule of writing electronic configuration :

The electronic configuration of any element is written in the ascending order of n. For example we can write the electronic configuration of Zr (Z = 40) following the Aufbau principle using Fig as follows :



But finally the above electronic configuration should be written according to value of n as follows:

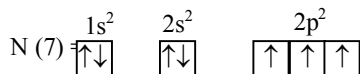
**3) Hund's Rule**

The distribution of electrons in different degenerate orbitals, i.e., orbitals of same energy, is governed by Hund's rule. **Hund's rule is as follows:**

The electrons will distribute themselves in different degenerate orbitals in such a way that maximum number of electrons remain in unpaired state. The spin of the unpaired electrons will be in the same direction. 'Degenerate orbitals' are three p-orbitals, five d-orbitals and seven f-orbitals. Hund's rule is not for s-orbital.

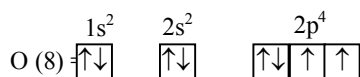
For example: The electronic configuration of N (7) = $1s^2 2s^2 2p^3$. In reality there are three 2p orbitals of same energy, which are designated as p_x , p_y and p_z

According to Hund's rule these last three electrons will remain in three different p-orbitals with their spin in same direction. So the electronic configuration of nitrogen N (7) in box system is as follows:



Here the arrow sign (↑) indicates the direction of spin of electrons.

Similarly the electronic configuration of oxygen O(8) = $1s^2 2s^2 2p^4$. More precisely O(8) = $1s^2 2s^2 2p_x^2 2p_y^1 2p_z^1$. Because the 4th electron in the 2p orbitals is bound to pair with one electron. The electronic configuration of oxygen atom in box system is as follows :

**Deviation of Electronic Configuration from the General Rule**

The electronic configuration of almost all elements can be confirmed according to Aufbau principle using Fig. 2.16. But there are some deviations. Two major reasons behind these deviations are as follows :

(1) **More stability of half filled and full filled orbitals :** It is found that the degenerate orbitals (p, d and f) gain more stability, when they are exactly half filled or full filled. That means that np^3 , np^6 , nd^5 , nd^{10} , nf^7 and nf^{14} configurations are more stable. For this reason the configuration $(n-1)d^5 ns^1$ becomes more stable than $(n-1)d^4 ns^2$ and $(n-1)d^9 ns^1$ becomes more stable than $(n-1)d^{10} ns^2$. Therefore the electronic configurations of Cr(24) and Cu(29) are as follows:

